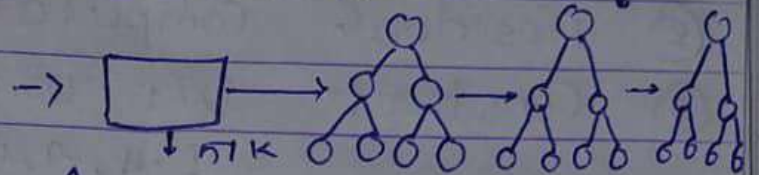


Xg boost Regressor ML Algorithm

Dataset & Regression y



$\rightarrow$ Exp	Gap	Salary	R_1	y	R_2
$\rightarrow$ 2	yes	40K	-11	49.9	-9.9
$\rightarrow$ 2.5	yes	42K	-9	49.9	-7.9
$\rightarrow$ 3	No	52K	1	51.5	0.5
4	No	60K	9	51.5	8.5
4.5	yes	62K	11	52.1	9.9

$\approx 51K$

Similar weight = $\frac{\sum (\text{Residual})^2}{\sum p \cdot (1-p)}$ use for classification.

Gain

$$[51 + (0.1)(-10)] = 51 - 0.1 = 49.9$$

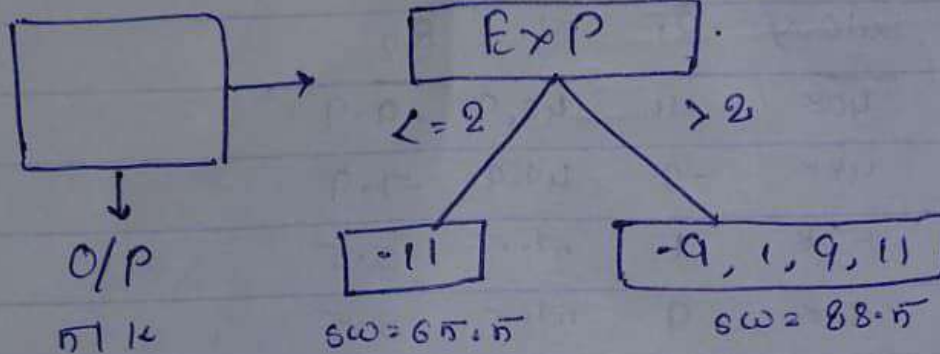
$$[51 + (0.1)(5)] = 51.5$$

$$[51 + (0.1)(11)] = 52.1$$

steps

- ① Create a Base Model
- ② Residual Computation
- ③ Construct DT1 using α, R, γ

$$[-11, -9, 1, 9, 11] \Rightarrow sw = 0.16$$



$$\text{similarity weight} = \frac{\sum (\text{Residual})^2}{\text{No. of Residuals} + \lambda} \rightarrow \lambda \text{ is hyperparameter}$$

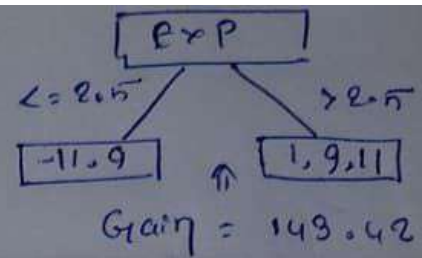
$\lambda = 1$

$$\boxed{\lambda \uparrow \text{ sw } \downarrow}$$

$$sw(\text{left child}) = \frac{121}{1+1} = \frac{121}{2} = 60.5$$

$$sw(\text{Right child}) = \frac{(-9+1+9+11)^2}{4+1}$$

$$= \frac{144}{5} = 28.8$$

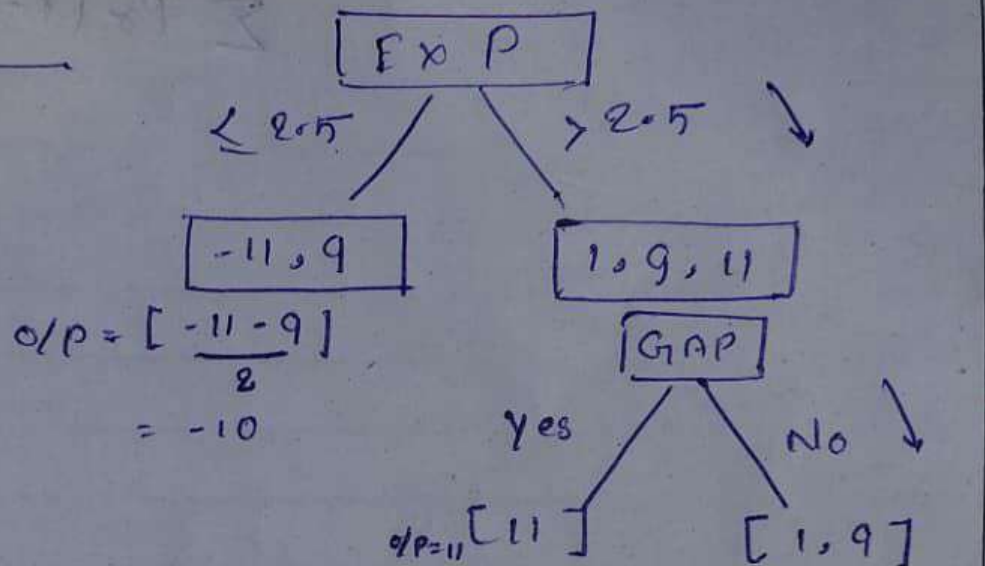
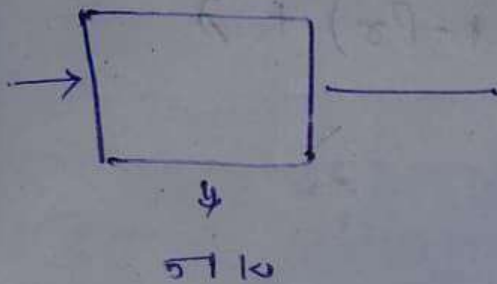


④ Calculate Gain

$$\text{Gain} = 65.5 + 22.5 - 0.16$$

$$= \underline{\underline{98.34}}$$

Base learner



α = Learning Rate

$\alpha = 0.1 \rightarrow$ Hyperparameter

$$\frac{1+9}{2} = 5$$

$$\text{XG Boost Classifier} = \text{Base learner} + \alpha_1(DT_1) + \alpha_2(DT_2) + \dots + \alpha_n(DT_n)$$

o/p

$$\begin{aligned} \text{XG Boost Regressor} &= 51K + 0.1(5) \\ &= 51 + 0.5 \\ &= 51.5 \end{aligned}$$

$$\text{Similarity weight} = \frac{\sum (\text{Residual})^2}{\text{No. of Residuals} + 1}$$

Regression

$$\text{Similarity weight} = \frac{\sum (\text{Residual})^2}{\sum p_x (1 - p_x) + 1}$$

